

Quiz 8

● Graded

Student

Samyak Sanghvi

Total Points

3 / 4 pts

Question 1

Q1

1 / 2 pts

✓ + 1 pt Correct definition of erosion

+ 2 pts Result = $\{9 \leq x \leq 19, 10 \leq y \leq 20\}$

+ 0 pts No/Wrong answer

Question 2

Q2

2 / 2 pts

✓ + 1 pt Opening: only big squares remain

✓ + 1 pt Closing of opening: bridge between squares returns

+ 0 pts No/Wrong answer

Name: Samyalk Sanghvi

COL7(6)83 Quiz 8

Entry no.: 2023CS1080>

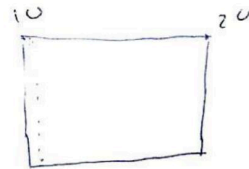
24 October, 2025

1. What is the effect of erosion with a structuring element $B = \{(1, 0)\}$, i.e. which contains only one pixel, just below the origin? Specifically, give the result $A \ominus B$ when A is the square $\{(x, y) : 10 \leq x \leq 20, 10 \leq y \leq 20\}$.

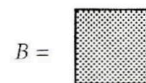
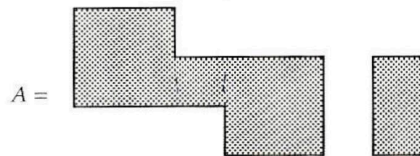
$$x, y \in A \ominus B \quad \text{iff} \quad (x+1, y) \in A$$

1 pixel layer from right is removed,

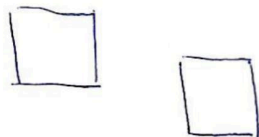
$$A \ominus B = \left\{ \begin{array}{l} 10 \leq x \leq 19 \\ 10 \leq y \leq 20 \end{array} \right\}$$



2. Sketch the opening $A \circ B$, and the closing of the result, $(A \circ B) \bullet B$, for the given sets A and B .

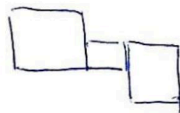


$$A \circ B =$$



(all regions $\subset B$ gone)

$$(A \circ B) \bullet B =$$



(all dark patches $\subset B$ gone)